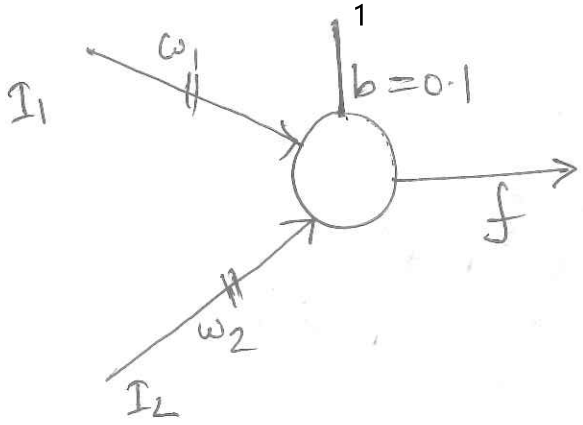
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Perceptron

A perceptron is a simple model of a biological neuron that is used in artificial neural networks. It is a machine-based algorithm that is used to learn how to classify data.

Q.



Training data

Input		output
I_1	I_2	d
1	1	1
0	0	0
0	1	?

initial weight

$\omega_2 = 0.3$

$\omega_1 = 0.5$

$b = 0.1$

Test data 0 1

find output for test data, learning rate $\eta = 0.1$.

Soln. for training data:

forward pass: pattern 1 ($I_1 = 1, I_2 = 1, d = 1$)

$$v = I_1 w_1 + I_2 w_2 + b$$
$$= 1 \times 0.5 + 1 \times 0.3 + 0.1$$
$$= 0.5 + 0.3 + 0.1$$
$$= 0.9$$

$$f = \frac{1}{1 + e^{-av}}$$

$$= \frac{1}{1 + e^{-0.9}} \quad ; [a = 1]$$

$$= 0.711$$

Backpropagation:

δ calculation:

$$\delta = f'(d-f)$$

$$\delta = f(1-f)(d-f)$$

$$= 0.711(1-0.711)(1-0.711)$$

$$= 0.05938$$

Weight update

$$\Delta w_1 = \eta * (I_1 * \delta)$$

$$= 0.1 * (1 * 0.05938)$$

$$= 0.005938$$

$$\therefore w_1 = \Delta w_1 + w_1$$

$$= 0.005938 + 0.5$$

$$= 0.5059$$

$$\Delta w_2 = \eta * (I_2 * \delta)$$

$$= 0.1 * (1 * 0.05938)$$

$$= 0.005938$$

$$\therefore w_2 = \Delta w_2 + w_2$$

$$= 0.005938 + 0.3$$

$$= 0.3059$$

$$\Delta b = \eta * (1 * 0.059)$$

$$= 0.1(1 * 0.059)$$

$$= 0.005938$$

$$b = b + \Delta b$$

$$= 0.1 + 0.00593*$$

$$= 0.1059$$

forward pass, pattern 2 ($I_1=0$, $I_2=0$, $d=0$)

$$v = I_1 w_1 + I_2 w_2 + b$$

$$= 0 \times 0.5059 + 0 \times 0.3059 + 0.1059$$

$$= 0.1059$$

$$\therefore f = \frac{1}{1+e^{-av}}$$

$$= \frac{1}{e^{-1059}+1} \quad ; [a=1]$$

$$= \frac{1}{1+0.8995}$$

$$= 0.526$$

Back propagation

δ calculation

$$\delta = f'(d-f)$$

$$= f(1-f)(d-f)$$

$$= 0.526(1-0.526)(0-0.526)$$

$$= -0.1311$$

weight update

$$\Delta w_1 = \eta * (I_1 * \delta)$$

$$= 0.1 * (0 * (-0.1311))$$

$$= 0$$

$$w_1 = w_1 + \Delta w_1$$

$$= 0.5059 + 0$$

$$= 0.5059$$

$$\Delta w_2 = \eta * (I_2 * \delta)$$

$$= 0.1 * (0 * (-0.1311))$$

$$= 0$$

$$w_2 = 0.3059 + 0$$

$$= 0.3059$$

$$\Delta b = \eta * (1 * (-0.1311))$$

$$= -0.01311$$

$$b = b + \Delta b$$

$$= 0.1059 + (-0.01311)$$

$$= 0.09279$$

Testing.

$$v = I_1 w_1 + I_2 w_2 + b$$

$$= 0 * 0.5059 + 1 * 0.3059 + 0.09279$$

$$= 0.39869$$

$$\therefore f = \frac{1}{1+e^{-av}}$$

$$= \frac{1}{1+e^{-0.39869}} \quad ; [a=1]$$

$$= \frac{1}{1+0.67} = \frac{1}{1.67} = 0.598$$ Ans: